Summary in Graph

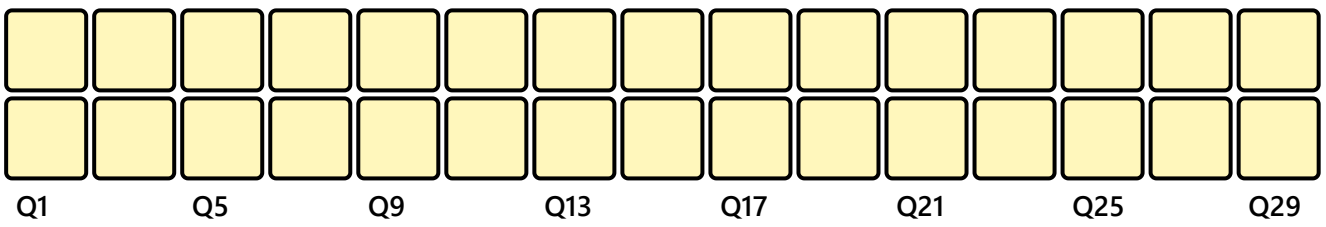
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (8) out of 81 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Mathematical Logic

Which of the following is the correct formalization for the sentence "No student loves Bill"?

(Mark all the appropriate choices)

A. $\neg\exists x(\text{Student}(x) \wedge \text{Loves}(x, Bill))$

B. $\forall x(\neg \text{Student}(x) \vee \neg \text{Loves}(x, Bill))$
C. $\forall x(\neg \text{Student}(x) \vee \text{Loves}(x, Bill))$
D. $\forall x(\neg \text{Student}(x) \wedge \neg \text{Loves}(x, Bill))$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Consider the following logical statements:

- P_1 : If it rained or one of the team did not turn up, the match would not be played.
- P_2 : The match was played.

Which of the following is not a valid conclusion?

A. It did not rain.
B. It did not rain and no team turned up
C. One team turned up
D. All are valid conclusions

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let's consider a propositional language where

- p means "x is a power of 2",
- q means "x is even".

Which of the following is the correct formal representation of the sentence "x being a power of 2 is a sufficient condition for x being even"?

A. $p \rightarrow q$
B. $q \rightarrow p$
C. $\neg p \rightarrow \neg q$
D. none of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Two finite sets X and Y have m and n elements each. The number of subsets of X is 96 more than that of Y. The values of m, n respectively are

A. $4, 7$
B. $7, 4$
C. $5, 7$
D. $7, 5$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

P and Q are two prepositions. Which of the following logical expressions are equivalent?

I. $\neg Q \rightarrow \neg P$
II. $\neg P \vee Q$
III. $P \rightarrow Q$
IV. $\neg P \wedge Q$

A. I, IV
B. I, II, III
C. II, IV
D. I, II, III, IV

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following is the negation of the statement
"All pentagon are polygon".

A. All pentagon are not polygon
B. There is some pentagon which is not polygon
C. All polygons are pentagon
D. There is some pentagon which is polygon

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Consider the following two statements about sets X, Y and Z
S_1 : If $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$.
S_2 : If $X \in Y$ and $Y \in Z$, then $X \in Z$

A. Only S_1 is true
B. Only S_2 is true
C. Both S_1 and S_2 are true
D. Neither S_1 nor S_2 is true

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Which of the following is a valid partition for N, where N is the set of natural numbers? (Mark all the appropriate choices)

A. $\{x \in N \mid x \geq 100\} \cup \{x \in N \mid x \leq 100\}$
B. $\{x \in N \mid x \geq 100\} \cup \{x \in N \mid x < 100\}$
C. $\{x \in N \mid x > 100\} \cup \{x \in N \mid x < 100\}$
D. $\{x \in N \mid x \geq 101\} \cup \{x \in N \mid x < 100\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Consider set $A = \{\phi, \{1,2\}, 4, \{3\}\}$. Which of the following is a subset of A? (Mark all the appropriate choices)

A. $\{1,2\}$
B. $\{\{3\}\}$
C. $\{3\}$
D. $\{4,3\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let A, B and C be three sets. Then $(A-B)-C$ is _____.

A. $(A-B)-(B-C)$
B. $(A-C)-(B-C)$
C. $(B-C)-(A-C)$
D. $(A-B)-(A-C)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Consider the following two identities on three sets X, Y and Z. (Here Δ represents the symmetric difference operation)

- $S_1 : X-(Y \cup Z) = (X-Y) \cap (X-Z)$
- $S_2 : X \Delta (Y \cap Z) = (X \Delta Y) \cap (X \Delta Z)$

Which of the following options is TRUE?

A. Only S_1 is True
B. Only S_2 is True
C. Both S_1 and S_2 are True
D. Neither S_1 nor S_2 is True

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let $a(t), b(t)$ and $c(t)$ be the lengths of three sides of a geometric figure t in non-increasing order (i.e., $a(t) \geq b(t) \geq c(t)$).
We define the sets:

■$X := \{t : a(t) = b(t)\}$
■$Y := \{t : b(t) = c(t)\}$
■$T := \{\text{the set of all triangles}\}$

Which of the following will represent an equilateral triangle?

A. $(X \cup Y) \cap T$
B. $X \cap Y \cap T$
C. $X \oplus Y \oplus T$
D. $X \cap Y$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Consider set $A = \{1, 2, 3, 4, 5\}$. From all the subsets of A, we define a new set X as $X = \{(P, Q) \mid |P \Delta Q| = 2\}$, where (P, Q) is ordered pair (P, Q) and Δ denotes the symmetric difference operation. The number of elements in X, i.e., $|X|$ is _____

Your Answer: | Correct Answer: 320 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Consider three finite sets A, B and C. Let $S_1 = ((A \cap B) - (B \cap C)) - C$ and $S_2 = (A - (A - C)) - (A - B)$. The sets S_1 and S_2 are related as.

A. $S_1 \subset S_2$
B. $S_1 \supset S_2$
C. $S_1 = S_2$
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 07sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Consider the following relations defined on set I:

- $R_1 : \{(x, y) \mid xy \geq 1\}$
- $R_2 : \{(x, y) \mid x \equiv y(\mod 7)\}$
- $R_3 : \{(x, y) \mid x \neq y\}$

where I is set of all non-negative integers. Which of the following statements is TRUE?

A. R_1 reflexive while R_2 and R_3 are not reflexive.
B. R_1 and R_3 are not reflexive while R_2 is reflexive.
C. R_1, R_2 and R_3 are all reflexive.
D. None of these relations is reflexive

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let $P(X)$ be the collection of all the subset of a set X with at least three elements. Consider the following two relations on $P(X)$:

- $R : A \subseteq B$
- $S : A \cap B = \phi$.

Which of the following is true about relations R and S?

A. R is anti-symmetric but S is not anti-symmetric.
B. R is not anti-symmetric but S is anti-symmetric.
C. Both R and S are anti-symmetric
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Consider the following two sentences

- S_1 : The intersection of two transitive relations is transitive.
- S_2 : The union of two transitive relations is transitive.

Which of the below options is/are CORRECT?

A. Only S_1
B. Only S_2
C. Neither S_1 nor S_2
D. Both S_1 and S_2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

If R and S are two equivalence relations on a set A, then which of the below statements is/are TRUE?

- S_1 : $R \cup S$ is an equivalence relation.
- S_2 : $R \cap S$ is an equivalence relation.

A. S_1 is true and S_2 is false
B. S_1 is false and S_2 is true
C. Neither S_1 nor S_2 is true
D. Both S_1 and S_2 are true

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Define a relation P on the set of all functions from $\mathbb{R}$ to $\mathbb{R}$ as
fPg if and only if there is a $k \in \mathbb{R}$ such that $f(x) = g(x)$ for every $x \geq k$. Which of the following is TRUE?

A. P is reflexive as well as symmetric
B. P is not reflexive
C. P is not Transitive
D. P is not Symmetric

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Given a relation $*$ on Z^2 by $(a, b) * (c, d)$ if and only if either $a < c$ or else $a = c$ and $b \leq d$. Consider the following statements.

- S_1 : $*$ is transitive.
- S_2 : $*$ is anti-symmetric.

A. S_1 is TRUE and S_2 is FALSE
B. S_1 is FALSE and S_2 is TRUE

C. Neither S_1 nor S_2 is correct
D. Both S_1 and S_2 are correct.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

The domain of the function f defined by $f(x) = \frac{1}{\sqrt{x+|x|}}$ is.

A. $\mathbb{R}$
B. $\mathbb{R}^+$
C. $\mathbb{R} - \{0\}$
D. None of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Consider the following function f on $\left[-\frac{1}{2}, 3\right] \to [-1, 1]$ such that $f(x) = \frac{2x-1}{4x+5}$. Which of the following statements is TRUE?

A. f is not one-one but is onto
B. f is neither one-one nor onto
C. f is bijective
D. f is one-one but not onto

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following is/are bijection from $\mathbb{R}$ to $\mathbb{R}$? (Mark all the appropriate choices)

A. $2x + 1$
B. $x^2 + 1$
C. x^3
D. x

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Suppose that $h : \mathbb{Z} \to \mathbb{Z}$ is known to be one-one and onto (bijection). Let's define a function $f : \mathbb{Z}^2 \to \mathbb{Z}$ by $f(x, y) = h(x) + h(y)$. Which of the following defines the nature of function $f(x, y)$?

A. f is one-one
B. f is onto
C. f is a bijection
D. f is neither one-one nor onto

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let f be a function from set A to set B. Let S and T be two subsets of A. Now, consider the statements given below.

- $S_1 : f(S \cup T) = f(S) \cup f(T)$.
- $S_2 : f(S \cap T) = f(S) \cap f(T)$.

Which of the above statements is/are TRUE?

A. Only S_1 is true
B. Only S_2 is true
C. Both S_1 and S_2 are true
D. Neither S_1 nor S_2 is true

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

For every athlete A, if A is successful then A has good skills and A is hardworking.
Which one of the following sentences is equivalent to the above claim.

A. For every athlete A, if A does not have good skills or A is not hardworking, then A is not successful
B. There is an athlete A, such that if A does not have good skills or A is not hardworking, then A is not successful
C. If A does not have good skills or A is not hardworking, then A is not successful
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Consider a propositional language where

- p means "Bonny is happy",
- q means "Bonny plays a music",
- r means "Seema is happy".

The propositional formula for the sentence:

"if Bonny is happy and plays a music then Seema isn't happy" is?

(Mark all the appropriate choices)

A. $p \wedge q \rightarrow \neg r$
B. $p \vee q \rightarrow \neg r$
C. $\neg p \vee \neg q \vee \neg r$
D. $p \vee q \rightarrow r$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Consider the following statements assuming there is at least one tomato, one fruit and one vegetable:

- S_1 : All tomatoes are vegetables. All fruits are vegetables. Therefore, some tomatoes are fruits.
- S_2 : Some tomatoes are fruits. All fruits are vegetables. Therefore, some tomatoes are vegetables.

Which of the following options is correct about the above statements?

A. S_1 is correct but S_2 is not correct
B. S_1 is not correct but S_2 is correct
C. Neither S_1 nor S_2 is correct.
D. Both S_1 and S_2 are correct

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Which of the following is correct formalization of the sentence

"There is a student who is loved by every other student."?

Here, $\text{Loves}(p, q)$ means p loves q.

A. $\exists x(\text{Student}(x) \land \forall y(\text{Student}(y) \rightarrow \text{Loves}(y, x)))$
B. $\exists x(\text{Student}(x) \land \forall y(\text{Student}(y) \land \neg(y = x) \rightarrow \text{Loves}(y, x)))$
C. $\exists x(\text{Student}(x) \rightarrow \forall y(\text{Student}(y) \land \text{Loves}(y, x)))$
D. $\exists x(\text{Student}(x) \rightarrow \forall y(\text{Student}(y) \land \neg(y = x) \land \text{Loves}(y, x)))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Match the pairs in the following question by choosing appropriate option.

P_1	A	There are exactly two mangoes
P_2	B	There are at least two mangoes
P_3	C	There is exactly one mango
P_4	D	There are at least three mangoes

where,
$P_1 : \exists x(\text{Mango}(x) \land \forall y(\text{Mango}(y) \implies y = x))$
$P_2 : \exists x \exists y(\text{Mango}(x) \land \text{Mango}(y) \land (x \neq y) \land \forall x \forall y \forall z((\text{Mango}(x) \land \text{Mango}(y) \land \text{Mango}(z)) \implies (x =$
$P_3 : \exists x \exists y(\text{Mango}(x) \land \text{Mango}(y) \land (x \neq y)$
$P_4 : \exists x \exists y \exists z(\text{Mango}(x) \land \text{Mango}(y) \land \text{Mango}(z) \land (x \neq y) \land (y \neq z) \land (x \neq z)$

A. $P_1 - C, P_2 - A, P_3 - D, P_4 - B$
B. $P_1 - C, P_2 - B, P_3 - D, P_4 - A$
C. $P_1 - C, P_2 - A, P_3 - B, P_4 - D$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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